GAO YUNZHE

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EDUCATION

Nanyang Technological University (NTU) Singapore
M.Sc. in Computer Control and Automation **Aug. 2025 – Jan. 2027 (Expected)**

- **GPA: 4.63 / 5.0**
- Honors: EEE Global Merit Award (Top 2% of the cohort; ∼20 recipients out of 1,000+ eligible students).
- Advisor: Prof. Mao Kezhi
- Dissertation Topic: Design and Deployment of Lightweight Large Language Models (LLM) for Efficient, Localized Deployment.
- Selected Coursework: Artificial Intelligence & Data Mining, Genetic Algorithms & Machine Learning, Computer Control Networks.

Northeastern University (NEU) Shenyang, China
B.E. in Automation **Sept. 2021 – Jul. 2025**

- **GPA: 3.76 / 5.0 (Avg. Score: 87.6/100)**
- Honors: University Academic Scholarship.
- Selected Coursework: Artificial Intelligence Foundation (93/100), Distributed Cooperative Control (96/100), Cloud Computing and Industrial Internet (87/100).

RESEARCH EXPERIENCE

Energy-Efficient Edge-Cloud Collaborative LLM Inference NTU, Singapore
Master Dissertation (Advisor: Prof. Mao Kezhi) Aug. 2025 – Present

- Proposed "Green-Edge-Cloud," an uncertainty-aware collaborative inference framework that reframes query routing as a model-intrinsic confidence calibration problem, enabling zero-shot decision-making via intrinsic confidence scores without auxiliary routing networks.
- Developed a routing-oriented CoT distillation pipeline using LoRA to transfer reasoning capabilities from a 72B teacher to a 1.5B edge student, explicitly aligning the student's confidence with actual correctness and mitigating overconfidence.
- Built a hardware-level energy profiling pipeline with `CodeCarbon`, explicitly modeling edge idle energy during cloud offloading to capture real-world latency–energy trade-offs beyond conventional FLOPs-based proxies.
- Achieved a strong accuracy–efficiency trade-off on GSM8K and SVAMP, recovering over 50% of the accuracy gap while reducing total energy consumption by **81.16%** and accelerating inference by **5.6**× compared to cloud-only baselines.

Democratizing Fine-Grained Manipulation via Imitation Learning NEU, China
Undergraduate Thesis (Advisor: Prof. Li Zhi) Nov. 2024 – Jun. 2025

- Developed a low-cost, 6-DOF mobile teleoperation system (Koch v1.1, under $250) integrating the Action Chunking Transformer (ACT) and Conditional VAE (CVAE) to mitigate compounding errors in continuous control.
- Replaced the standard ResNet backbone with MobileNetV2 and integrated YOLOv5 for dynamic object segmentation, reducing inference latency to 0.014s (20% faster) and enabling real-time 60Hz control.
- Successfully deployed the framework in real-world scenarios, achieving an 86% grasping success rate in high-precision plug insertion tasks and a 96.3% success rate in dynamic logistics sorting tasks.

Coal Ash Composition Prediction via Machine Learning NEU, China
Project Lead (First Author) Jan. 2024 – Aug. 2024

- Proposed a novel inversion method using Particle Swarm Optimization - Extreme Learning Machine (PSO-ELM) to predict coal ash content from Near-Infrared (NIR) spectral data.

- Designed a hybrid feature selection pipeline combining Savitzky-Golay (SG) smoothing and Random Frog (RF) algorithms, effectively reducing spectral redundancy.
- Achieved state-of-the-art prediction accuracy with an R^2 of 0.98 and RMSE of 0.79, validated on real-world industrial datasets.
- Outcomes: Published findings as the First Author in Control Engineering of China (Core Journal).

Digital Twin-based Mineral Processing Intelligent Decision System NEU, China
Core Algorithm Engineer Jan. 2023 – Nov. 2023

- Engineered the "Abnormal Pattern Retrieval System (ECAS)" using Digital Twin technology to monitor massive industrial equipment data.
- Implemented correlation analysis algorithms to retrieve abnormal operational patterns, significantly enhancing fault detection efficiency.
- Outcomes: Awarded the Grand Prize at the NEU Innovation and Entrepreneurship Competition, the National Silver Award at the "Internet+" Competition (Top 1% nationwide), and secured 2 Software Copyrights.

PUBLICATIONS

[1] **Y. Gao**, J. Zou, D. Xiao. "Prediction of Coal Ash Composition Based on Near-Infrared Visible Spectroscopy and PSO-ELM." *Control Engineering of China*, 2024. (**Core Journal**)

[2] **Y. Gao**, Z. Feng, K. Mao. "Green-Edge-Cloud: Routing without Routers via Confidence Calibration." *ICML 2026 Workshop (EIML)*. **Accepted**.

INTELLECTUAL PROPERTY

[1] **Mineral Processing Equipment Abnormal Pattern Retrieval and Index Correlation Analysis System (ECAS) V1.0**
K. Zhang, **Y. Gao** (2nd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1469715, Nov. 2023

[2] **Intelligent Venue Resource Allocation and Security Operation Visual Analysis Software (IVVAS) V1.0**
K. Zhang, P. Zhu, **Y. Gao** (3rd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1421636, Nov. 2023

SELECTED AWARDS

- **Global Merit Award**, Nanyang Technological University 2025-2026
- **National Silver Award (Top 1% nationwide)** Dec. 2023
  China International College Students' Innovation Competition
- **University Academic Scholarship**, Northeastern University 2024-2025
- **University Academic Scholarship**, Northeastern University 2021-2022

TECHNICAL SKILLS

- **Programming & Languages:** Python, C/C++, MATLAB, SQL, LaTeX.
- **AI & LLM Frameworks:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, TRL), vLLM, CUDA, CodeCarbon, Scikit-learn.
- **Robotics & Simulation:** ROS (Robot Operating System), MuJoCo, Gazebo, MobileNet/YOLO.
- **Systems & Deployment:** Linux, Docker, Git, Edge Computing Optimization.
- **Core Algorithms:** LLM Quantization & Knowledge Distillation, Action Chunking Transformer (ACT), Reinforcement Learning, PSO-ELM (Swarm Intelligence), Digital Twin.